

## EXPERIMENT-2

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**Subject Name:** Computer Networks

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**AIM:** Study the basic network command and Network configuration commands like ping, variations of ipconfig, tracert, nslookup, netstat, arp, hostname, pathping and basic networking commands.

- ping
- ipconfig
- tracert
- nslookup
- netstat
- arp
- hostname
- pathping
- route
- net use

## OBJECTIVES:

- To study the basic Windows networking commands used for network configuration and troubleshooting.
- To understand the working of the Ping command and its variations for testing network connectivity using ICMP.
- To learn the use of IPConfig commands for viewing and managing TCP/IP configuration.

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- To understand the Tracert command for tracing the route taken by packets from the source to the destination.
- To study the NSLookup command for DNS name resolution and reverse lookup.
- To examine active network connections, listening ports, and protocol statistics using the Netstat command.
- To understand the ARP command for viewing and managing the ARP cache (IP-to-MAC address mapping).
- To learn the use of the Hostname, PathPing, Route, and Net Use commands for identifying the system, analyzing network paths, viewing routing information, etc.

### SOFTWARE REQUIREMENTS:

#### **(a) Software:**

- Microsoft Windows 11 Operating System
- Windows Command Prompt (CMD)

#### **(b) Hardware (Virtual):**

- Desktop Computer or Laptop
- Network Interface Card (NIC)
- Internet/LAN Connection

### **THEORY:** (i) Ping

**Ping (Packet Internet Groper)** is a network diagnostic utility used to verify whether a remote host or network device is reachable over an IP network. It works by sending **ICMP (Internet Control Message Protocol) Echo Request** packets to the destination host and waiting for **ICMP Echo Reply** packets. If the destination responds, the command displays the round-trip time (RTT), packet loss, and Time To Live (TTL) value. Ping is one of the most commonly used tools for testing network connectivity and troubleshooting communication problems.

#### **Working of Ping**

1. The source computer sends an ICMP Echo Request packet to the destination.
2. The destination receives the packet and sends back an ICMP Echo Reply.
3. The source measures the time taken for the packet to travel to the destination and back.

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4. If no reply is received within a specified time, the request is considered timed out.

### **Purpose of Ping**

- To verify whether a host is reachable.
- To test network connectivity.
- To measure response time (latency).
- To detect packet loss.
- To troubleshoot network problems.

### **Common Variations**

- ping <IP/hostname> – Tests connectivity.
- ping -t – Sends continuous Echo Requests until stopped.
- ping -n <count> – Sends a specified number of Echo Requests.
- ping -l <size> – Specifies the size of the data payload.
- ping -f – Sets the "Don't Fragment" flag.
- ping -a – Resolves an IP address into its hostname.

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### **(ii) IPConfig**

**IPConfig (Internet Protocol Configuration)** is a Windows command-line utility used to display and manage the TCP/IP configuration of network interfaces. It provides information such as the IP address, subnet mask, default gateway, DHCP status, and DNS server details. It is an essential command for verifying network configuration and troubleshooting IP-related issues.

### **Purpose**

- Display network configuration.
- Verify IP address assignment.
- Check subnet mask and gateway.

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- Renew or release DHCP addresses.
  - Troubleshoot IP connectivity problems.
- 

### (iii) Tracert

**Tracert (Trace Route)** is a Windows networking command that traces the complete path taken by packets from the source computer to the destination. Unlike Ping, which only confirms whether a destination is reachable, Tracert identifies every intermediate router (hop) through which the packets travel. It uses ICMP Echo Request packets with increasing TTL (Time-To-Live) values.

#### **Working**

- Sends packets with TTL = 1.
- The first router decreases TTL to zero and returns an ICMP Time Exceeded message.
- TTL is gradually increased to discover each router along the path.
- The process continues until the destination is reached.

#### **Purpose**

- Display the routing path.
  - Identify slow routers.
  - Detect routing failures.
  - Troubleshoot Internet connectivity.
- 

### (iv) NSLookup

**NSLookup (Name Server Lookup)** is a command-line utility used to query the Domain Name System (DNS). DNS translates human-readable domain names into IP addresses. NSLookup helps verify DNS functionality and diagnose DNS-related issues.

#### **Purpose**

- Resolve domain names to IP addresses.

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- Perform reverse DNS lookup.
  - Verify DNS server operation.
  - Retrieve different DNS record types.
- 

### (v) Netstat

**Netstat (Network Statistics)** is a command-line utility that displays network-related information such as active connections, listening ports, routing tables, protocol statistics, and Ethernet statistics. It is widely used by network administrators to monitor network activity and identify suspicious or unnecessary connections.

#### **Purpose**

- View active TCP/UDP connections.
  - Display listening ports.
  - Show routing information.
  - Display protocol statistics.
  - Identify process IDs using network ports.
- 

### (vi) ARP

**ARP (Address Resolution Protocol)** is used to map an IPv4 address to its corresponding MAC address on a local network. Communication over Ethernet requires MAC addresses, while applications use IP addresses. ARP bridges this gap by maintaining an **ARP Cache**.

#### **Working**

1. The sender knows the destination IP.
2. It broadcasts an ARP Request.
3. The destination replies with its MAC address.
4. The sender stores the mapping in the ARP cache.

#### **Purpose**

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- View ARP cache.
  - Delete ARP entries.
  - Create static ARP entries.
  - Troubleshoot LAN communication.
- 

### (vii) Hostname

The **Hostname** command displays the name assigned to the current computer. Every computer connected to a network has a unique hostname for easy identification.

#### **Purpose**

- Identify the computer on the network.
  - Assist in network administration.
  - Verify the system name.
- 

### (viii) PathPing

**PathPing** combines the features of **Ping** and **Tracert**. It traces the complete path to the destination and then sends multiple ICMP Echo Requests to each router to calculate packet loss and network latency at every hop.

#### **Purpose**

- Detect packet loss.
- Identify slow routers.
- Analyze network latency.
- Troubleshoot network performance.

#### **Working**

1. Finds the route like Tracert.
2. Sends multiple ping packets to every hop.

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3. Calculates packet loss percentage.
  4. Displays latency and delay.
- 

### (ix) Route

The **Route** command is used to display and manage the routing table in Windows. A routing table contains information about the paths used to forward packets to different networks.

#### **Purpose**

- Display routing table.
  - Add static routes.
  - Delete routes.
  - Modify routing entries.
- 

### (x) Net Use

**Net Use** is a Windows networking command used to connect, disconnect, and manage shared network resources such as shared folders and printers. It allows users to map a network share to a drive letter for easier access.

#### **Purpose**

- Display shared network resources.
- Map network drives.
- Disconnect mapped drives.
- Manage network connections.

## STEPS/IMPLEMENTATION:

### Part I – Ping Command

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1. Open the Command Prompt (CMD) by pressing Windows + R, typing cmd, and pressing Enter.
2. Execute the Ping command to check network connectivity.
3. Observe the reply time, TTL value, and packet loss.
4. Execute different Ping variations such as -t, -n, -l, -f, and -a to understand their functions.

### Part II – IPConfig Command

6. Execute the ipconfig command to view the current IP configuration.
7. Execute ipconfig /all to display detailed TCP/IP configuration.
8. Execute ipconfig /release to release the current DHCP-assigned IP address.
9. Execute ipconfig /renew to obtain a new IP address from the DHCP server.
10. Observe the IP address, subnet mask, default gateway, DNS server, and MAC address.

### Part III – Tracert Command

11. Execute the tracert command to trace the route to a destination.
12. Observe the number of hops, router IP addresses, and response time for each hop.
13. Execute tracert -d to display only IP addresses.
14. Execute tracert -h to specify the maximum number of hops.

### Part IV – NSLookup Command

15. Execute the nslookup command to resolve a domain name into an IP address.
16. Perform reverse DNS lookup using an IP address.
17. Query different DNS record types such as **A** and **AAAA** records.
18. Verify DNS server functionality.



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### Part V – Netstat Command

19. Execute the netstat command to display active network connections.
20. Use netstat -a to display all active and listening ports.
21. Execute netstat -n, netstat -o, netstat -e, netstat -r, netstat -s, and netstat -ano.
22. Observe protocol statistics, routing information, Ethernet statistics, and Process IDs (PID).

### Part VI – ARP Command

23. Execute the arp -a command to display the ARP cache.
24. Observe the IP-to-MAC address mappings.
25. Delete ARP entries using arp -d.
26. Create a static ARP entry using arp -s.

### Part VII – Hostname Command

27. Execute the hostname command.
28. Observe the current computer name used on the network.

### Part VIII – PathPing Command

29. Execute the pathping command for a remote host.
30. Observe the complete route, packet loss percentage, and latency at each router.
31. Identify any router causing delays or packet loss.

### Part IX – Route Command

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32. Execute the route print command to display the routing table.
33. Add a static route using the route add command.
34. Delete the static route using the route delete command.
35. Observe the changes in the routing table.

### Part X – Net Use Command

36. Execute the net use command to display currently connected network resources.
37. Map a shared folder to a drive letter using the net use command.
38. Disconnect the mapped drive using net use /delete.
39. Verify that the network resource has been removed successfully.

## DEMONSTRATION/OUTPUT:

### Part-1: Ping command

```
C:\Users\USER>ping google.com
```

```
Pinging google.com [172.217.26.46] with 32 bytes of data:  
Reply from 172.217.26.46: bytes=32 time=225ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=274ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=283ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=383ms TTL=113
```

```
Ping statistics for 172.217.26.46:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 225ms, Maximum = 383ms, Average = 291ms
```

```
C:\Users\USER>ping -t google.com
```

```
Pinging google.com [172.217.26.46] with 32 bytes of data:  
Reply from 172.217.26.46: bytes=32 time=286ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=93ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=175ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=386ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=120ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=104ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=309ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=405ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=200ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=318ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=148ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=120ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=460ms TTL=113  
Reply from 172.217.26.46: bytes=32 time=165ms TTL=113
```

```
Ping statistics for 172.217.26.46:  
    Packets: Sent = 14, Received = 14, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 93ms, Maximum = 460ms, Average = 234ms
```

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```
C:\Users\USER>ping -n 5 8.8.8.8
```

```
Pinging 8.8.8.8 with 32 bytes of data:
```

```
Reply from 8.8.8.8: bytes=32 time=310ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=121ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=134ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=211ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=158ms TTL=114
```

```
Ping statistics for 8.8.8.8:
```

```
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 121ms, Maximum = 310ms, Average = 186ms
```

```
C:\Users\USER>ping -a 8.8.8.8
```

```
Pinging dns.google [8.8.8.8] with 32 bytes of data:
```

```
Reply from 8.8.8.8: bytes=32 time=404ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=89ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=176ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=444ms TTL=114
```

```
Ping statistics for 8.8.8.8:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 89ms, Maximum = 444ms, Average = 278ms
```

```
C:\Users\USER>ping -f -l 138 8.8.8.8
```

```
Pinging 8.8.8.8 with 138 bytes of data:
```

```
Reply from 8.8.8.8: bytes=138 time=302ms TTL=114  
Reply from 8.8.8.8: bytes=138 time=315ms TTL=114  
Reply from 8.8.8.8: bytes=138 time=332ms TTL=114  
Reply from 8.8.8.8: bytes=138 time=425ms TTL=114
```

```
Ping statistics for 8.8.8.8:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 302ms, Maximum = 425ms, Average = 343ms
```

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```
C:\Users\USER>ping -l 10 8.8.8.8
```

```
Pinging 8.8.8.8 with 10 bytes of data:
```

```
Reply from 8.8.8.8: bytes=10 time=174ms TTL=117
```

```
Reply from 8.8.8.8: bytes=10 time=156ms TTL=117
```

```
Reply from 8.8.8.8: bytes=10 time=400ms TTL=117
```

```
Reply from 8.8.8.8: bytes=10 time=402ms TTL=117
```

```
Ping statistics for 8.8.8.8:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 156ms, Maximum = 402ms, Average = 283ms
```

### Part-2: IPConfig Command

```
C:\Users\USER>ipconfig/release
```

```
Windows IP Configuration
```

```
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
```

```
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
```

```
No operation can be performed on Bluetooth Network Connection while it has its media disconnected.
```

```
Wireless LAN adapter Local Area Connection* 1:
```

```
Media State . . . . . : Media disconnected
```

```
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Local Area Connection* 2:
```

```
Media State . . . . . : Media disconnected
```

```
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix . :
```

```
Link-local IPv6 Address . . . . . : fe80::e5fc:e681:f98f:a24d%3
```

```
Default Gateway . . . . . :
```

```
Ethernet adapter VMware Network Adapter VMnet1:
```

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```
C:\Users\USER>ipconfig/all
```

### Windows IP Configuration

```
Host Name . . . . . : DESKTOP-A0I2075
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : cuchdit.in
```

### Wireless LAN adapter Local Area Connection\* 1:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 92-E8-68-D5-CF-21
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

### Wireless LAN adapter Local Area Connection\* 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
#2
Physical Address. . . . . : 92-E8-68-D5-CF-31
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

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```
C:\Users\USER>ipconfig
```

```
Windows IP Configuration
```

```
Wireless LAN adapter Local Area Connection* 1:
```

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
```

```
Wireless LAN adapter Local Area Connection* 2:
```

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix  . :
```

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix  . : cuchdit.in
Link-local IPv6 Address . . . . . : fe80::e5fc:e681:f98f:a24d%3
IPv4 Address. . . . . : 172.25.187.75
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . : 172.25.176.1
```

```
Ethernet adapter VMware Network Adapter VMnet1:
```

```
Connection-specific DNS Suffix  . :
Link-local IPv6 Address . . . . . : fe80::3f12:532a:2f9e:245e%15
IPv4 Address. . . . . : 192.168.111.1
Subnet Mask . . . . . : 255.255.255.0
```

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```
C:\Users\USER>ipconfig /renew
```

```
Windows IP Configuration
```

```
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
```

```
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.
```

```
No operation can be performed on Bluetooth Network Connection while it has its media disconnected.
```

```
Wireless LAN adapter Local Area Connection* 1:
```

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Local Area Connection* 2:
```

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Wi-Fi:
```

```
Connection-specific DNS Suffix . : cuchdit.in  
Link-local IPv6 Address . . . . . : fe80::e5fc:e681:f98f:a24d%3  
IPv4 Address. . . . . : 172.25.190.29  
Subnet Mask . . . . . : 255.255.240.0  
Default Gateway . . . . . : 172.25.176.1
```

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### Part-3: Tracert Command

```
C:\Users\USER>tracert 8.8.8.8
```

```
Tracing route to dns.google [8.8.8.8]  
over a maximum of 30 hops:
```

|    |        |        |        |                      |
|----|--------|--------|--------|----------------------|
| 1  | 4 ms   | 6 ms   | 3 ms   | 172.25.176.1         |
| 2  | 3 ms   | 3 ms   | 2 ms   | 192.168.50.11        |
| 3  | 6 ms   | 2 ms   | 2 ms   | 192.168.50.1         |
| 4  | 5 ms   | 2 ms   | 4 ms   | 172.16.2.1           |
| 5  | 137 ms | 100 ms | 126 ms | 47.247.118.29        |
| 6  | *      | *      | *      | Request timed out.   |
| 7  | 252 ms | 131 ms | 128 ms | 142.251.201.128      |
| 8  | 215 ms | 127 ms | 281 ms | 192.178.83.245       |
| 9  | 54 ms  | 41 ms  | 42 ms  | 209.85.252.65        |
| 10 | 181 ms | 208 ms | 175 ms | dns.google [8.8.8.8] |

```
Trace complete.
```

```
C:\Users\USER>tracert -d 8.8.8.8
```

```
Tracing route to 8.8.8.8 over a maximum of 30 hops
```

|    |        |       |       |                    |
|----|--------|-------|-------|--------------------|
| 1  | 7 ms   | 4 ms  | 2 ms  | 172.25.176.1       |
| 2  | 4 ms   | 4 ms  | 2 ms  | 192.168.50.11      |
| 3  | 5 ms   | 3 ms  | 2 ms  | 192.168.50.1       |
| 4  | 3 ms   | 3 ms  | 5 ms  | 172.16.2.1         |
| 5  | 106 ms | 83 ms | 89 ms | 49.249.148.145     |
| 6  | *      | *     | *     | Request timed out. |
| 7  | *      | *     | *     | Request timed out. |
| 8  | *      | *     | *     | Request timed out. |
| 9  | *      | *     | *     | Request timed out. |
| 10 | 21 ms  | 39 ms | 41 ms | 72.14.196.97       |
| 11 | 54 ms  | 43 ms | 43 ms | 142.251.77.189     |
| 12 | 44 ms  | 41 ms | 42 ms | 142.251.52.205     |
| 13 | 50 ms  | 87 ms | 83 ms | 8.8.8.8            |



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```
C:\Users\USER>tracert -h 5 8.8.8.8
```

```
Tracing route to dns.google [8.8.8.8]  
over a maximum of 5 hops:
```

```
  1    28 ms    3 ms    5 ms  172.25.176.1  
  2     3 ms    3 ms    2 ms  192.168.50.11  
  3    23 ms    4 ms    2 ms  192.168.50.1  
  4     3 ms    1 ms    2 ms  172.16.2.1  
  5    18 ms   45 ms   38 ms  dsl.delhi.209.17.246.61.airtelbroadband.in [61  
.246.17.209]
```

```
C:\Users\USER>tracert -?
```

```
Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]  
             [-R] [-S srcaddr] [-4] [-6] target_name
```

Options:

|                 |                                                 |
|-----------------|-------------------------------------------------|
| -d              | Do not resolve addresses to hostnames.          |
| -h maximum_hops | Maximum number of hops to search for target.    |
| -j host-list    | Loose source route along host-list (IPv4-only). |
| -w timeout      | Wait timeout milliseconds for each reply.       |
| -R              | Trace round-trip path (IPv6-only).              |
| -S srcaddr      | Source address to use (IPv6-only).              |
| -4              | Force using IPv4.                               |
| -6              | Force using IPv6.                               |

### Part-4: NSLookup Command

```
C:\Users\USER>nslookup -type=A google.com  
Server:  DC2K16.cuchdit.in  
Address:  172.19.2.101
```

Non-authoritative answer:

```
Name:      google.com  
Address:   172.217.26.46
```

```
C:\Users\USER>nslookup google.com  
Server:  DC2K16.cuchdit.in  
Address:  172.19.2.101
```

Non-authoritative answer:

```
Name:      google.com  
Addresses: 2404:6800:4002:830::200e  
          172.217.26.46
```

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### Part-5: Netstat Command

```
C:\Users\USER>netstat -e
Interface Statistics
```

|                     | Received  | Sent     |
|---------------------|-----------|----------|
| Bytes               | 480866370 | 49506841 |
| Unicast packets     | 372600    | 129738   |
| Non-unicast packets | 0         | 2161     |
| Discards            | 0         | 0        |
| Errors              | 0         | 0        |
| Unknown protocols   | 0         |          |

```
C:\Users\USER>netstat -r
```

```
=====
Interface List
 9...92 e8 68 d5 cf 21 .....Microsoft Wi-Fi Direct Virtual Adapter
16...92 e8 68 d5 cf 31 .....Microsoft Wi-Fi Direct Virtual Adapter #2
 3...90 e8 68 d5 cf 61 .....MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
15...00 50 56 c0 00 01 .....VMware Virtual Ethernet Adapter for VMnet1
 6...00 50 56 c0 00 08 .....VMware Virtual Ethernet Adapter for VMnet8
12...90 e8 68 d5 cf 60 .....Bluetooth Device (Personal Area Network)
 1.....Software Loopback Interface 1
=====
```

#### IPv4 Route Table

```
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          172.25.176.1     172.25.190.29    30
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
172.25.176.0                255.255.240.0    On-link          172.25.190.29    286
172.25.190.29              255.255.255.255  On-link          172.25.190.29    286
172.25.191.255             255.255.255.255  On-link          172.25.190.29    286
192.168.111.0              255.255.255.0    On-link          192.168.111.1    291
192.168.111.1              255.255.255.255  On-link          192.168.111.1    291
192.168.111.255            255.255.255.255  On-link          192.168.111.1    291
192.168.217.0              255.255.255.0    On-link          192.168.217.1    291
192.168.217.1              255.255.255.255  On-link          192.168.217.1    291
192.168.217.255            255.255.255.255  On-link          192.168.217.1    291
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
=====
```

## EXPERIMENT-2

```
C:\Users\USER>netstat -o
```

### Active Connections

| Proto | Local Address       | Foreign Address               | State       | PID   |
|-------|---------------------|-------------------------------|-------------|-------|
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:62232         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:62240         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:62243         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:62245         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:62250         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:62252         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:62253         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:62257         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62232     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62237     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62240     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62243     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62245     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62247     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62248     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62250     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62252     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62253     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62255     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 127.0.0.1:62257     | DESKTOP-A0I2075:49671         | TIME_WAIT   | 0     |
| TCP   | 172.25.190.29:49426 | 4.213.25.242:https            | ESTABLISHED | 6076  |
| TCP   | 172.25.190.29:50490 | whatsapp-cdn-shv-01-del3:5222 | ESTABLISHED | 23364 |
| TCP   | 172.25.190.29:51179 | 20.249.115.161:https          | ESTABLISHED | 11020 |
| TCP   | 172.25.190.29:51568 | 116.119.124.163:https         | CLOSE_WAIT  | 23364 |
| TCP   | 172.25.190.29:51596 | 4.207.44.75:https             | TIME_WAIT   | 0     |
| TCP   | 172.25.190.29:52660 | relay-03dda133:6568           | ESTABLISHED | 5272  |
| TCP   | 172.25.190.29:55944 | 23.103.238.165:https          | ESTABLISHED | 11020 |
| TCP   | 172.25.190.29:58651 | 52.110.14.159:https           | ESTABLISHED | 10364 |
| TCP   | 172.25.190.29:59204 | 52.123.253.73:https           | ESTABLISHED | 11020 |
| TCP   | 172.25.190.29:59754 | lb-140-82-112-26-iad:https    | ESTABLISHED | 19648 |

```
C:\Users\USER>netstat
```

### Active Connections

| Proto | Local Address       | Foreign Address               | State       |
|-------|---------------------|-------------------------------|-------------|
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:51578         | TIME_WAIT   |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:51579         | TIME_WAIT   |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:51584         | TIME_WAIT   |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:51590         | TIME_WAIT   |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:51597         | TIME_WAIT   |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:51598         | TIME_WAIT   |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:51599         | TIME_WAIT   |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:51601         | TIME_WAIT   |
| TCP   | 127.0.0.1:49671     | DESKTOP-A0I2075:51602         | TIME_WAIT   |
| TCP   | 127.0.0.1:51578     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51579     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51584     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51590     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51592     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51593     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51597     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51598     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51599     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51601     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51602     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 127.0.0.1:51603     | DESKTOP-A0I2075:49671         | TIME_WAIT   |
| TCP   | 172.25.190.29:49426 | 4.213.25.242:https            | ESTABLISHED |
| TCP   | 172.25.190.29:50490 | whatsapp-cdn-shv-01-del3:5222 | ESTABLISHED |
| TCP   | 172.25.190.29:51179 | 20.249.115.161:https          | ESTABLISHED |
| TCP   | 172.25.190.29:51568 | 116.119.124.163:https         | CLOSE_WAIT  |
| TCP   | 172.25.190.29:51569 | 4.207.44.75:https             | TIME_WAIT   |
| TCP   | 172.25.190.29:51576 | 20.184.175.12:https           | ESTABLISHED |
| TCP   | 172.25.190.29:51586 | 52.110.14.162:https           | TIME_WAIT   |
| TCP   | 172.25.190.29:51587 | 52.110.14.162:https           | TIME_WAIT   |
| TCP   | 172.25.190.29:51588 | 52.110.14.162:https           | TIME_WAIT   |
| TCP   | 172.25.190.29:51594 | 52.110.14.171:https           | TIME_WAIT   |
| TCP   | 172.25.190.29:51596 | 4.207.44.75:https             | ESTABLISHED |

## EXPERIMENT-2

### Part-6: ARP Command

```
C:\Users\USER>arp -a
```

|                                  |                   |         |
|----------------------------------|-------------------|---------|
| Interface: 172.25.190.29 --- 0x3 |                   |         |
| Internet Address                 | Physical Address  | Type    |
| 172.25.176.1                     | cc-5a-53-d3-d5-68 | dynamic |
| 172.25.191.255                   | ff-ff-ff-ff-ff-ff | static  |
| 224.0.0.2                        | 01-00-5e-00-00-02 | static  |
| 224.0.0.22                       | 01-00-5e-00-00-16 | static  |
| 224.0.0.251                      | 01-00-5e-00-00-fb | static  |
| 224.0.0.252                      | 01-00-5e-00-00-fc | static  |
| 239.255.102.18                   | 01-00-5e-7f-66-12 | static  |
| 239.255.255.250                  | 01-00-5e-7f-ff-fa | static  |
| 255.255.255.255                  | ff-ff-ff-ff-ff-ff | static  |

  

|                                  |                   |        |
|----------------------------------|-------------------|--------|
| Interface: 192.168.217.1 --- 0x6 |                   |        |
| Internet Address                 | Physical Address  | Type   |
| 192.168.217.255                  | ff-ff-ff-ff-ff-ff | static |
| 224.0.0.22                       | 01-00-5e-00-00-16 | static |
| 224.0.0.251                      | 01-00-5e-00-00-fb | static |
| 224.0.0.252                      | 01-00-5e-00-00-fc | static |
| 239.255.255.250                  | 01-00-5e-7f-ff-fa | static |

  

|                                  |                   |        |
|----------------------------------|-------------------|--------|
| Interface: 192.168.111.1 --- 0xf |                   |        |
| Internet Address                 | Physical Address  | Type   |
| 192.168.111.255                  | ff-ff-ff-ff-ff-ff | static |
| 224.0.0.22                       | 01-00-5e-00-00-16 | static |
| 224.0.0.251                      | 01-00-5e-00-00-fb | static |
| 224.0.0.252                      | 01-00-5e-00-00-fc | static |
| 239.255.255.250                  | 01-00-5e-7f-ff-fa | static |

### Part-7: Hostname

```
C:\Users\USER>hostname  
DESKTOP-A0I2075
```

## EXPERIMENT-2

### Part-8: Pathping Command

```
C:\Users\USER>pathping google.com
```

```
Tracing route to google.com [142.250.146.139]
```

```
over a maximum of 30 hops:
```

```
 0  DESKTOP-A0I2075.cuchdit.in [172.25.190.29]
 1  172.25.176.1
 2  192.168.50.11
 3  192.168.50.1
 4  172.16.2.1
 5  dsl.delhi.209.17.246.61.airtelbroadband.in [61.246.17.209]
 6  182.79.152.79
 7  142.250.161.56
 8  142.251.76.107
 9  216.239.62.218
10  192.178.124.241
11  192.178.254.238
12  142.251.251.55
13  172.253.50.210
14  *           *           *
```

```
Computing statistics for 325 seconds...
```

## EXPERIMENT-2

### Part-9: Route Command

```
C:\Users\USER>route print
```

```
=====
Interface List
 9...92 e8 68 d5 cf 21 .....Microsoft Wi-Fi Direct Virtual Adapter
16...92 e8 68 d5 cf 31 .....Microsoft Wi-Fi Direct Virtual Adapter #2
 3...90 e8 68 d5 cf 61 .....MediaTek Wi-Fi 6 MT7921 Wireless LAN Card
15...00 50 56 c0 00 01 .....VMware Virtual Ethernet Adapter for VMnet1
 6...00 50 56 c0 00 08 .....VMware Virtual Ethernet Adapter for VMnet8
12...90 e8 68 d5 cf 60 .....Bluetooth Device (Personal Area Network)
 1.....Software Loopback Interface 1
=====
```

### IPv4 Route Table

```
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          172.25.176.1     172.25.190.29    30
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
172.25.176.0               255.255.240.0    On-link          172.25.190.29    286
172.25.190.29              255.255.255.255  On-link          172.25.190.29    286
172.25.191.255             255.255.255.255  On-link          172.25.190.29    286
192.168.111.0              255.255.255.0    On-link          192.168.111.1    291
192.168.111.1              255.255.255.255  On-link          192.168.111.1    291
192.168.111.255            255.255.255.255  On-link          192.168.111.1    291
192.168.217.0              255.255.255.0    On-link          192.168.217.1    291
192.168.217.1              255.255.255.255  On-link          192.168.217.1    291
192.168.217.255            255.255.255.255  On-link          192.168.217.1    291
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link          172.25.190.29    286
224.0.0.0                  240.0.0.0        On-link          192.168.217.1    291
224.0.0.0                  240.0.0.0        On-link          192.168.111.1    291
255.255.255.255            255.255.255.255  On-link          127.0.0.1        331
255.255.255.255            255.255.255.255  On-link          172.25.190.29    286
255.255.255.255            255.255.255.255  On-link          192.168.217.1    291
=====
```

### Part-10: Net use Command

```
C:\Users\USER>net use
New connections will be remembered.

There are no entries in the list.
```

## EXPERIMENT-2

### LEARNING OUTCOMES:

After completing this experiment:

- I understood the purpose and working of essential Windows networking commands such as Ping, IPConfig, Tracert, NSLookup, Netstat, ARP, Hostname, PathPing, Route, and Net Use, and learned how they are used for network configuration, monitoring, and troubleshooting.
- I gained practical knowledge of network connectivity testing by using the Ping command and its various options to verify communication between devices, measure network latency, identify packet loss, and diagnose basic network connectivity issues.
- I learned to view and manage TCP/IP network configuration using IPConfig, trace packet routes using Tracert, resolve domain names using NSLookup, and analyze routing paths and packet loss using PathPing, enabling effective diagnosis of network communication problems.
- I developed the ability to monitor and analyze network activity using Netstat, understand IP-to-MAC address mapping through ARP, identify the system using the Hostname command, and manage routing information and shared network resources using the Route and Net Use commands.
- I enhanced my practical skills in network administration and troubleshooting by executing command-line networking utilities in Windows, interpreting their outputs, identifying common network issues, and applying appropriate commands to verify connectivity, resolve configuration problems, and manage network resources efficiently.